Chapter	2
Unit	10-16
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unit 10 Q1

```
prime_list = [2, 3, 5, 7]

print("first element of prime_list:"  ,prime_list[0])
```

unit 10 Q2

```
prime_list = [2, 3, 5, 7]

print("Prime numbers before append:", prime_list)

prime_list.append(11)

print("Prime numbers after append:", prime_list)
```

unit 10 Q3

```
list1 = [3,5,7]

list2 = [ 2,3,4,5,6]

for i in list1:

   for j in list2:

      print(f"{i} * {j} = {i*j}")
```

unit 11 Q1

```
capitals_dic =  {"Korea":"Seoul", "China":"Beijing", "USA":"Washington D.C"}

print(capitals_dic["Korea"])
```

unit 11 Q2

```
fruits_dic = {"apple":"5000KRW", "banana":"4000KRW", "grape":"5300KRW", "melon":"6500KRW"}

for key in fruits_dic:

   print(f"The price of {key} is {fruits_dic[key]}.")
```

unit 12 Q1

t1 = "a", "b", "c"

t2 = ("a", "b", "c")

t3 =("d","e")

print(t1==t2) -> True print(t1>t3) -> False print(t1<t3)->True

print(t2+t3)-> ('a', 'b', 'c', 'd', 'e')

print([t2+t3])-> [('a', 'b', 'c', 'd', 'e')]

print(t1)-> ('a', 'b', 'c')

unit 12 Q2

```
dailySale = (100,121,120,130,140,120,122,123,190,125)
count = 0
for i in range(len(dailySale)-1):
    if dailySale[i+1] < dailySale[i]:
        count += 1
print(f"In the last 10 days, {count} days had reduced sales compared to previous day.")
```

unit 13 Q1

```
list_array = [[10,20],[30,40],[50,60]]
print(list_array[1][0])
```

unit 13 Q2

```
twoD_array = [[1,2,3,4],[5,6,7,8],[9,10,11,12],[13,14,15,16]]
for i in range(4):
    for j in range(4):
        print(twoD_array[i][j], end = " ")
    print()
```

unit 14 Q1

```
person_dic = {"Last Name":"Doe", "First Name":"David","Company":"Samsung"}

for key in person_dic:
    print(f"{key}: {person_dic[key]}")
```

unit 14 Q2

```
items = { "Coffee":7, "Pen":3, "Milk":1,"Coke":4,"Book":5}

itemsAsked = input("Enter the item you want to check: ")

if itemsAsked in items:
    print(f"number of {itemsAsked} in stock is {items[itemsAsked]}")
```

unit 15 Q1

```
student_tup = (('211101' ,'David Doe' , '010-1234-4500') , ('211102' ,'John Smith','010-2230-6540') , ('211103' ,'Jane Carter', '010-3232-7788'))

studentDict = {}

for i in student_tup:
    student ={}
    student[i[0]] = list(i[1:])
    print(student)
    studentDict.update(student)
```

unit 15 Q2

```
Id = input("Enter the student ID: ")

if Id in studentDict:
    print(f"Name: {studentDict[Id][0]}")
    print(f"Phone Number: {studentDict[Id][1]}")
```

unit 16 Q1

```
lst = ["apple", "mango", "banana"]
s1 = set(lst)
print(s1)
```

unit 16 Q2

```
s1 = {10,20,30,40}
s2 = {30,40,50,60,70}
print(s1|s2)
{70, 40, 10, 50, 20, 60, 30}
print(s1&s2)
{40, 30}
print(s1-s2)
{10, 20}
print(s1^s2)
{50, 20, 70, 10, 60}
print(s1.issubset(s2))
False
print(s1.issuperset(s2))
False
print(s1.isdisjoint(s2))
False
```